MID-TERM EXAMINATION

(Programme: B.Tech. CSE/ECE-AI/ECE/IT/MAE/DMAM) (Semester: II)

(March 2026) OFF LINE mode

Subject Code: BAS 104	**Subject: Environmental Sciences**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**
	a) Explain the remedial steps taken (with chemical reactions) for removal of nutrients from industrial waste water.	
	b) Discuss the principle of determination of Chemical Oxygen Demand (COD) in water with suitable reactions.	
	c) How is carbon dioxide sequestered artificially?	
	d) Calculate the atom Economy of Sodium benzoate as desired product in the following reaction: $C_6H_5COOC_2H_5+NaOH \rightarrow C_6H_5COONa+C_2H_5OH$	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) 20 ml of standard hard water (containing 15 g of $CaCO_3$ per L) required 25 mL EDTA solution for end-point. 100 mL of water sample required 18 mL EDTA solution; while the same water after boiling required 12 mL EDTA solution. Calculate carbonate and non-carbonate hardness of water. Which buffer is used in this titration and what is its pH?	
	b) A sample of water was alkaline to both phenolphthalein and methyl orange. 50 mL of this water sample required 6 mL of N/50 HCl for phenolphthalein end point and 9 mL of the acid to methyl orange end point. Determine the types and extent of alkalinity present.	
	c) How are chloramines formed during disinfection process of water for domestic use and what is their significance? Discuss.	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Discuss the chemical treatment methods to treat hazardous waste?	
	b) Discuss the sources, sinks, effects and control of particulate matter as air pollutant.	
	c) Discuss the merits of green method over traditional method for synthesis of adipic acid, with suitable reactions.	